CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:
Ship Date: DEC 4th
Destination: Cincinnati
Recipient: Sue Smith
Phone #: 513-569-7631
Contact:
Phone #:
E-Mail:

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required
20171127						DI	FIELDBLK		TN02, TN02-3, TNH4, TRP, TP, TN
20171129	10:06:00	FL	02	0.1	1	SW	UNK		"
20171129	10:30:00	FL	05	0.1	1	SW	UNK		"
20171129	11:09:00	FL	01	0.1	1	SW	UNK		"
20171129	11:40:00	FL	07	0.1	1	SW	UNK		"
20171129	12:03:00	FL	15	0.1	1	SW	UNK		"
20171129	12:31:00	FL	14	0.1	1	SW	UNK		"
20171129	12:52:00	FL	03	0.1	1	SW	UNK		"
20171129	13:21:00	FL	04	0.1	1	SW	UNK		"
20171129	13:58:00	FL	45	0.1	1	SW	UNK		"
20171129	14:25:00	FL	49	0.1	1	SW	UNK		"
20171129	15:19:00	FL	43	0.1	1	SW	UNK		"
20171129	15:44:00	FL	47	0.1	1	SW	UNK		"
20171130	12:00:00				1	DI	HOUSE BLK		"
20171127					1	DI	Field BLK	2MHCl	TOC
20171129	10:06:00	FL	02	0.1	1	SW	UNK	2MHCl	TOC
20171129	10:30:00	FL	05	0.1	1	SW	UNK	2MHCl	TOC
20171129	11:09:00	FL	01	0.1	1	SW	UNK	2MHCl	TOC
20171129	11:40:00	FL	07	0.1	1	SW	UNK	2MHCl	TOC
20171129	12:03:00	FL	15	0.1	1	SW	UNK	2MHCl	TOC
20171129	12:31:00	FL	14	0.1	1	SW	UNK	2MHCl	TOC
20171129	12:52:00	FL	03	0.1	1	SW	UNK	2MHCl	TOC
20171129	13:21:00	FL	04	0.1	1	SW	UNK	2MHCl	TOC
20171129	13:58:00	FL	45	0.1	1	SW	UNK	2MHCl	TOC
20171129	14:25:00	FL	49	0.1	1	SW	UNK	2MHCl	TOC
20171129	15:19:00	FL	43	0.1	1	SW	UNK	2MHCl	TOC
20171129	15:44:00	FL	47	0.1	1	SW	UNK	2MHCl	TOC
20171130	12:00:00				1	DI	HOUSE BLK	2MHCl	TOC

Relinquished By:	Date:	Time:	Received By:	Date:	Time:

CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:
Ship Date: Dec 4th
Destination: Cincinnati, OH
Recipient: Sam Smith
Phone #: 513-569-7681

Contact:
Phone #:
E-Mail:

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)	
20171129		FL	02	0.1	1	SW	UNK	Chloro	306.31	
20171129		FL	05	0.1	1	SW	UNK	Chloro	338.35	
20171129		FL	01	0.1	1	SW	UNK	Chloro	294.45	
20171129		FL	07	0.1	1	SW	UNK	Chloro	286.34	
20171129		FL	15	0.1	1	SW	UNK	Chloro	297.28	
20171129		FL	14	0.1	1	SW	UNK	Chloro	250.37	
20171129		FL	03	0.1	1	SW	UNK	Chloro	200.55	
20171129		FL	04	0.1	1	SW	UNK	Chloro	188.98	
20171129		FL	45	0.1	1	SW	UNK	Chloro	180.83	
20171129		FL	49	0.1	1	SW	UNK	Chloro	121.96	
20171129		FL	43	0.1	1	SW	UNK	Chloro	102.35	
20171129.		FL	47	0.1	1	SW	UNK	Chloro	147.64	
20171127					1	DI	Field BLK	Chloro	326.97	

Relinquished By:	Date:	Time:	Received By:

GAS TRAP DATA SHEET

— = 0 = none collected

Lake:						
Site	**Lat**	**Long**	**Deployment time/date**	**Retrieval time/date**	**Volume recovered (mL)**	**Exetainer codes**
24	35.942055	78.588155	11/28 10:35	11/29 9:41	1mL	
16	35.956693	78.590421	10:45	9:52	—	
Dir 2 ✓	35.976428	78.588715	11:00	10:06	—	
5 ✓	35.970225	78.621261	11:10	10:30	2mL	
12	35.965533	78.633951	11:20	10.40	8mL	
13	36.003691	78.646355	11:34	10:58	—	
1 ✓	36.002090	78.634448	11:41	11:09	—	
21	36.009035	78.628178	11:48	11:21	—	
7 ✓	36.005603	78.667986	12:00	11:40	8mL	112817.FL.07.BT.1;2;3
11	36.019301	78.717040	12:27	12:12	2mL	
15 ✓	36.022840	78.714623	12:30	12:03	—	
14 ✓	34.059648	78.717680	12:50	12:31	—	
3 ✓	36.032466	78.736816	13:04	12:52	3mL	
8	36.018920	78.748686	13:11	13:08	20mL	112817.FL.08.BT.1
4 ✓	36.012743	78.760618	13:16	13:21	8mL	
10	36.039766	78.756018	13:35	13:41	—	
45 ✓	36.044106	78.758168	13:40	13:58	—	
57	36.048590	78.752138	13:52	14:14	—	
49 ✓	36.051268	78.756908	13:58	14:25	2mL	
55	36.053256	78.761733	14:00	14:45	2mL	
61	36.053465	78.759478	14:04	14:36	22	112817.FL.61.BT.1
63	36.056115	78.770190	14:26	15:07	1mL	
54	36.057908	78.765226	14:33	14:57	3mL	Trap poss. broke
Dir 43 ✓	36.061503	78.773226	14:41	15:19	15	112817.FL.43.BT.1
58	36.068606	78.775995	14:50	15:32	5mL	
47 ✓	36.068511	78.768513	14:57	11/29 15:44	8mL	

DISSOLVED GAS AND AIR DATA SHEET

Lake:	Falls Lake				Date: 11/29/17			
¥Site	Dissolved gas Exetainer codes		Water Vol/ Helium Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes			BP (mm Hg)
2	11282017.FL.02.	DG.1 : 2		15.3	11282017.FL.02.	AA.1:2:3:	4:5	
5	11282017.FL.	05.DG.1 : 2		15.9				
1	11282017.FL.	01.DG.1 : 2		14.9				
7	11282017.FL.	07.DG.1 : 2		18.0				
15	11282017.FL.	15.DG.1 : 2		15.4				
14	11282017.FL.	14.DG.1 : 2		16.3				
3	11282017.FL.	03.DG.1 : 2		18.0				
4	11282017.FL.	04.DG.1 : 2		15.6				
45	11282017.FL.	45.DG.1 : 2		18.3				
49	11282017.FL.	49.DG.1 : 2		16.8			Two samples in 1 vial	Lots of bird around
43	11282017.FL.	43.DG.1 : 2		16.9	11282017.FL.	43AA.1:2:3:	4:5	759.7.
47	11282017.FL.	47.DG.1 : 2		16.0				

3 ; 7 are switched

¥Site: all dissolved gas samples collected at 0.1m depth

Boat motor should be off for 2 minutes before sample collection

Goal is to inject 20mL of gas into exetainer

Signature Date

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES	
Lake:	Falls Lake
Sample Date:	11/29/17 – 11/30/17
GGA file	gga-2017-11-29_f0000.txt gga-2017-11-30_f0000.txt
gps file name	

11/29 Time on pad [redacted] time on GPS Pro [redacted]

DEPLOYMENT DATA 11/30 9:17:35 14:19:00

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L)	CH_4 and CO_2 profile checked	
24	09:40:00	N	Y	50	Y	
16	09:52:37	N	Y	50	Y	
2	10:06:50	N	Y	50	Y	
5	10:30:52	Y	Y	50	Y	
12	10:40:50	N	Y	50	Y	
13	10:59:10	N	Y	50	Y	
1	11:10:14	N	Y	50	Y	
21	11:22:15	N	Y	55	Y	
7	11:41:00	N	Y	55	Y	
15	12:03:57	N	Y	55	Y	
11	12:13:50		Y	50	Y	First 2min good
14	12:30:59	N	Y	55	Y	Sandy area, bumpy
3	12:53:35	N	Y	50	Y	
8	13:09:42	N	Y	50	Y	
4	13:22:00	N	Y	50	Y	
10	13:41:23	N	Y	50	Y	Sandy
45	13:58 14:00:33	N	Y	40	Y	Near road, yards away
57	14:14:59	N	Y	50	Y	
49	14:27:42	N	Y	50	Y	
61	14:37:38	N	Y	45	Y	
55	14:47:00	N	Y	50	Y	
54	14:57:44	N	Y	55	Y	
63	15:07:37	N	Y	55	Y	
43	15:19:33	N	Y	50	Y	
58	15:31:00	N	Y	55	Y	
47	15:45:23	N	Y	50	Y	
52	9:10:21	N	Y	50	Y	
56	9:26:00	N	Y	50	Y	
68	9:38.26	N	Y	50	Y	

11/30

SONDE DATA SHEET

Lake: FALLS Sample Date: 11/28/17 – 11/30/17

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
24		0.1	13.07	~~8.93~~	~~91.9~~	0.082	6.24	~~5.6~~ 6.7	3.5	11/28
16		0.1	12.9	8.43	79.8	0.083	6.56	10.2	4.4	11/28
2	3.4 m	0.1	12.23	10.74	100.2	0.083	7.12	9.2	4.7	11/29
5	7.0 m	0.1	12.6	9.6	90.0	0.083	7.18	12.3	4.3	11/29
12	3.5	0.1	12.6	10.08	94.7	0.083	7.18	8.3	3.7	11/29
13	4.7	0.1	12.37	10.25	95.1	0.084	7.23	5.2	4.9	11/29
1	2.7	0.1	12.38	10.72	100.1	0.084	7.21	8.5	3.8	11/29
21	2.8	0.1	11.83	11.04	101.8	0.084	7.22	8.9	3.7	11/29
7	6.4 surface	0.1	12.06	10.15	94.3	0.088	7.25	7.3	4.7	11/29
7	6.4	6m	11.27	9.43	86.1	0.089	7.22	9.3	8.9	11/29
15	3.6 m	0.1	12.65	10.65	100.2	0.10	7.25	5.9	7.0	11/29
11	2.5	0.1	12.53	10.42	97.7	0.100	7.26	5.1	5.2	11/29
14	0.7	0.1	11.74	10.93	100.8	0.098	7.31	5.7	5.6	11/29
3	0.8	0.1	12.47	11.1	104.2	0.119	7.37	4.6	7.6	11/29
8	3.1 m	0.1	11.5	10.8	99.2	0.123	7.41	7.3	5.8	11/29
4	0.6	0.1	11.6	11.06	101.9	0.123	7.39	8.0	6.8	11/29
10	0.8	0.1	12.1	11.05	102.7	0.155	7.43	6.6	6.8	11/29
57	0.8	0.1	11.75	11.06	102.0	.172	7.52	8.0	9.6	11/29
49	1.8	0.1	11.63	11.16	103.5	.201	7.53	15.9	10.5	11/29
61	1.5	0.1	12.75	10.85	102.1	.206	7.63	14.6	12.0	11/29
55	1.6 m	0.1	12.52	11.34	107.0	0.212	7.67	72.5	12.5	11/29
54	0.8 m	0.1	12.31	11.8	110.8	0.221	7.8	17.9	14.9	11/29
63	1.5	0.1	11.67	11.66	108.5	0.211	7.92	12.7	15.2	11/29
43	0.8	0.1	12.15	12.45	115.8	0.234	8.05	18.1	21.5	11/29
58	0.8	0.1	12.27	11.38	106.3	0.247	8.17	21.3	13.8	11/29 BM

759.7 mmHg

	Site depth	Sample depth	Temp 15	DO mg/L	DO % sat	Sp. Cond.	pH	Chl a	Turbidity	Notes
47	1.3	0.1	~~12.86~~	11.76	112.6	.214	8.23	12.0	11.5	11/29
52	0.6	0.1	10.23	11.29	100.6	.212	8.18	12.5	17.6	11/30
~~50~~			~~10.09~~	~~11.62~~	~~103.3~~	~~.213~~	~~8.0~~	~~9.1~~	~~8.5~~	
56	1.4	0.1	10.16	11.6	102.8	0.215	7.96	11.3	11.3	11/30
68	1.5	0.1	10.05	10.96	97.3	0.217	7.91	14.0	12.5	11/30

Signatu

11/30/17

Date

Chamber LGR

Date 11/27/17
Cal cylinder serial # CC163816
Dilution system serial # TEI A20808 EPA #
LGR file 999_2017-11-27_f0000.ext

Zero Air Flow (Lpm)	Cal Gas Flow Set (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time	
3.25	0.0	0.0	0.0	0.13	7.24	9:37:30	9:43:00	Zero 20.6 LPM
9	7 ~~9.4~~	2.1	20.9	2.31	23.13	10:08:20 ~~9:58:00~~	10:13:20 ~~10:02~~	
5.3	7 ~~9.4~~	3.6	35.5	3.95	39.4	10:14:30	10:19:55	
5.3	10 ~~13.4~~	5.1	50.7	5.45	54.45	10:20:23	10:26:00	
3.25	15 ~~20.2~~	12.4	124.1	12.9	129.0	10:27:06	10:32:00	
3.25	20 ~~26.9~~	16.5	165.5	16.9	170.0	10:32:33	10:38:16	
3.25	~~30~~ ~~40.3~~	24.8	248.3	24.9	252.0	10:39:00	10:44:32	
3.25	60 ~~80.7~~	49.6	496.5	47.1	488.0	10:45:15	10:50:20	
3.25	70 ~~94.1~~	57.9	579.3	54.2	566.0	10:50:46	10:56:15	
3.25	90 ~~121.0~~	74.4	744.8	68.0	720.0	10:56:45	11:01:59	
3.25	0	0	0	0.014	0.31	9:44:43	09:50:00	

Signature _____

chamber LGR

Date 11/30/17 - Post Sampling
Cal cylinder serial #
Dilution system serial # A20808
LGR file 959_2017-11-30_f0001.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.034	0	14:33:00	14:37:00
9	7.0	2.1	20.9	2.38	23.74	14:37:00	14:39:45
5.3	7.0	3.6	35.5	4.03	40.30	14:~~48:20~~	14:53:20
5.3	10.0	5.1	50.7	5.53	55.3	14:53:20	14:54:20
3.25	15.0	12.4	124.1	13.08	132.5	14:54:20	14:56:25
3.25	20.0	16.5	165.5				
3.25	30.0	24.8	248.3				
3.25	60.0	49.6	496.5				
3.25	70.0	57.9	579.3				
3.25	90.0	74.4	744.8				

Signature _____